

Healthy People 2030 (in 2026), Early Childhood Development and Education

Literature Summary & References | Source: odphp.health.gov

About This Literature Summary

This summary of the literature on Early Childhood Development and Education as a social determinant of health is a narrowly defined examination that is not intended to be exhaustive and may not address all dimensions of the issue. Please note: The terminology used in each summary is consistent with the respective references. For additional information on cross-cutting topics, please see the [Language and Literacy](#) literature summary.

Related Objectives (6)

Here's a snapshot of the objectives related to topics covered in this literature summary. [Browse all objectives](#) .

- [Increase the proportion of 4th-graders with reading skills at or above the proficient level — AH-05](#)
- [Increase the proportion of 4th-graders with math skills at or above the proficient level — AH-06](#)
- [Increase the proportion of children and adolescents who communicate positively with their parents — EMC-01](#)
- [Increase the proportion of children whose family read to them at least 4 days per week — EMC-02](#)
- [Increase the proportion of children who are developmentally ready for school — EMC-D01](#)
- [Increase the proportion of children who participate in high-quality early childhood education programs — EMC-D03](#)

Related Evidence-Based Resources (4)

Here's a snapshot of the evidence-based resources related to topics covered in this literature summary. [Browse all evidence-based resources](#) .

- [Social Determinants of Health: Center-Based Early Childhood Education](#)

- Social Determinants of Health: Full Day Kindergarten Programs
- Social Determinants of Health: Out-of-School-Time Academic Programs – General
- Social Determinants of Health: School-Based Health Centers



Healthy People 2030 organizes the social determinants of health into 5 domains:

1. Economic Stability

2. Education Access and Quality
3. Health Care Access and Quality
4. Neighborhood and Built Environment
5. Social and Community Context

Literature Summary

Early childhood, particularly the first 5 years of life, impacts long-term social, cognitive, emotional, and physical development. ¹ , ² Healthy development in early childhood helps prepare children for the educational experiences of kindergarten and beyond. ³ Early childhood development and education opportunities are affected by various environmental and social factors, including early life stress, socioeconomic status, relationships with parents and caregivers, and access to early education programs. ⁴ , ⁵ , ⁶ , ⁷

Early life stress and adverse events can have a lasting impact on the mental and physical health of children. ³ Specifically, early life stress can contribute to developmental delays and poor health outcomes in the future. ³ , ⁴ , ⁸ , ⁹ Stressors such as physical abuse, family instability, unsafe neighborhoods, and poverty can cause children to have inadequate coping skills, difficulty regulating emotions, and reduced social functioning compared to other children their age. ³ , ⁴ , ⁹

Additionally, exposure to environmental hazards, such as lead in the home, can negatively affect a child's health and cause cognitive developmental delays. ³ Research shows that lead exposure disproportionately affects children from racial/ethnic minority and low-income households and can adversely affect their readiness for school. ¹⁰ , ¹¹

The socioeconomic status of young children's families and communities also significantly affects their educational outcomes. ⁶ Specifically, poverty has been shown to negatively influence the academic achievement of young children. ⁶ Research shows that, in their later years, children from disadvantaged backgrounds are more likely to repeat grades and drop out of high school. ¹ Children from communities with higher socioeconomic status and more resources experience safer and more supportive environments and better early education programs. ¹

Early childhood programs are a critical outlet for fostering the mental and physical development of young children. ² Some indicators of a high-quality early childhood development and education program include highly educated teachers, smaller classes, and lower child-staff ratios. ⁷ , ¹² High-quality early childhood programs can increase earning potential and encourage and support educational attainment. ¹³

Early childhood development and education programs can also help reduce educational gaps. ^{7 , 14 , 15} For example, Head Start is a federally funded early childhood program that provides comprehensive services for children from low-income families. ¹⁵ Head Start aims to improve health outcomes, increase learning and social skills, and close the gap in readiness to learn for children from low-income families and at-risk children. ¹⁵ Enrolling children in full-day kindergarten after the completion of preschool has also been shown to improve academic achievement. ¹⁴

Furthermore, extended early childhood programs for children up to 3rd grade, also referred to as booster programs, can provide comprehensive educational, health, and social services to complement standard early childhood and kindergarten programs. ^{14 , 16} These programs help sustain and bolster early developmental and academic gains. ^{14 , 16} Characteristics of such programs include: ¹⁴

- Low student-teacher ratio
- Focus on basic skills
- Teacher training
- Creation of school-parent liaisons
- School meals
- Provision of transportation to and from school
- Night courses for parents
- Health care services and referrals
- Home visitation
- Supportive social services

Quality education in elementary school is necessary to reinforce early childhood interventions and prevent their positive effects from fading over time. ¹⁴ Research also shows that the quality, length, and intensity of early education programs has an impact on well-being, including physical and mental health. ¹⁷ For example, children who enroll in low-quality schools with limited health resources, safety concerns, and low teacher support are more likely to have poorer physical and mental health. ^{18 , 19 , 20 , 21}

The developmental and educational opportunities that children have access to in their early years have a lasting impact on their health as adults. ^{13 , 22 , 23} The Carolina Abecedarian Project found that the children in the study who participated in a high-quality and comprehensive early childhood education program that included health care and nutritional components were in better health than those who did not. ²² The study found that, at age 21 years, the people who participated in the comprehensive early education program exhibited

fewer risky health behaviors; for example, they were less likely to binge drink alcohol, smoke cigarettes, and use illegal drugs.²² This group also self-reported better health and had a lower number of deaths.²²

Furthermore, by their mid-30s, the children who participated in the comprehensive early childhood development and education program had a lower risk for heart disease and associated risk factors, including obesity, high blood pressure, elevated blood sugar, and high cholesterol.¹³ These studies show that quality early childhood development and education programs can play a key role in reducing risky health behaviors and preventing or delaying the onset of chronic disease in adulthood.^{13 , 22}

Early childhood development and education are key determinants of future health and well-being.^{14 , 24} Addressing the disparities in access to early childhood development and education opportunities can greatly bolster young children's future health outcomes.^{9 , 14 , 15 , 22 , 23 , 25 - 31}

Additional research is needed to increase the evidence base for what can successfully impact the effects of childhood development and education on health outcomes and disparities. This additional evidence will facilitate public health efforts to address early childhood development and education as social determinants of health.

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